

Xuanrui Feng

Email: pkufengxuanrui@stu.pku.edu.cn

Homepage: fengxuanrui.github.io

Department: School of Mathematical Sciences (SMS)

School: Peking University (PKU)

Beijing, China

Education

Age	24
2024–2029 *	Ph.D., Analysis and PDE, Pure Mathematics, Peking University Advisor: Prof. Zhenfu Wang
2020–2024	B.Sc. Mathematics and Applied Mathematics, Peking University

Awards & Honors

2026	Zhong Jiaqing Outstanding Paper Award
2025	May-Fourth Scholarship of Peking University
2024	Outstanding Graduates of Beijing Regular Higher Education Institutions
2024	Outstanding Graduation Thesis of Beijing Universities
2024	Huaxin Bachelor at Beijing International Center for Mathematical Research
2023	National Scholarship
2022	National Scholarship

Research

 [Google Scholar](#)

Publications

- J1. Carrillo, J. A., Feng, X., Guo, S., Jabin, P.-E. & Wang, Z. Relative entropy method for particle approximation of the Landau equation for Maxwellian molecules. *Journal de Mathématiques Pures et Appliquées* 206, 103838 (2026).
- J2. Feng, X. & Wang, Z. Kac's Program for the Landau Equation. *Archive for Rational Mechanics and Analysis* 250, 72 (2026).
- J3. Feng, X. & Wang, Z. Quantitative propagation of chaos for 2D viscous vortex model on the whole space. *Peking Mathematical Journal*, 1–31 (2026).
- J4. Feng, X. & Wang, Z. Quantitative propagation of chaos for 2D viscous vortex model with general circulations on the whole space. *Nonlinearity* 39, 015009 (2026).

*Expected.

Preprints

- P1. Feng, X., Liu, C. & Wang, Z. Uniform-in-time relative entropy estimates for Kac's approximation of the Landau equation. arXiv preprint arXiv:2608.29815 (2026).
- P2. Feng, X. & Wang, Z. Modulated Gibbs measure and mean-field limit for 3D Vlasov–Poisson–Fokker–Planck equations. arXiv preprint arXiv:2609.08640 (2026).
- P3. Cai, S., Feng, X., Gong, Y. & Wang, Z. Propagation of chaos for 2D log gas on the whole space. arXiv preprint arXiv:2411.14777 (2024).

Talks

Talks

- T1. Kac's program for the Landau equation. 27th Beijing–Tianjin–Hebei Young Scholar Academic Conference on Probability and Statistics (Beijing Institute of Technology, Beijing, China). May 2026.
- T2. Kac's program for the Landau equation. Sino–Russian Student Mathematical Seminar (Peking University, Beijing, China). Apr. 2026.
- T3. Kac's program for the Landau equation. 3rd National Mathematics Frontier PhD Academic Exchange Forum (Peking University, Beijing, China). Nov. 2025.
- T4. Kac's program for the Landau equation. 23rd Annual Conference of China Society for Industrial and Applied Mathematics (CSIAM) (Changsha, China). Oct. 2025.
- T5. Kac's program for the Landau equation. Wuhan University Academic Talks (Wuhan, China). Oct. 2025.
- T6. Quantitative propagation of chaos for singular first-order system on the whole space. Workshop on Mean-Field Limit (Peking University, Beijing, China). Apr. 2025.
- T7. Quantitative propagation of chaos for singular first-order system on the whole space. Stochastic Webinar (Chinese Academy of Sciences and Beijing Institute of Technology, Online). Dec. 2024.

Teaching

Peking University

2025–2026	TA, Advanced Mathematics D
2024–2025	TA, Advanced Mathematics D

Last updated: September 18, 2026